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Assessment Alignment and Statistical Moderation of the 2021–2023 WASSCE Chemistry Multiple-Choice Examination in Akwa Ibom State, Nigeria

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Abstract

Validity, fairness, and alignment of high-stakes examinations are central to educational accountability systems. This study investigates the alignment of the 2021–2023 West African Senior School Certificate Examination (WASSCE) Chemistry multiple-choice items with the prescribed Nigerian curriculum and evaluates the neutrality of the regression-based statistical moderation process applied to School-Based Assessment (SBA) scores. A descriptive design was employed. Chemistry objective papers from 2021 to 2023 were analyzed using Bloom's Revised Taxonomy, Item-Objective Congruence (IOC) indices, and a curriculum mapping matrix. Additionally, 585 anonymized student score records from 24 secondary schools in Akwa Ibom State were analyzed using paired-sample and independent-sample t-tests to assess the impact of regression moderation. Results indicated substantial coverage of topics across core curriculum domains, although item distribution was uneven, with a greater concentration on chemical reactions and periodicity. IOC indices demonstrated moderate alignment (mean IOC > 0.56). However, 96% of items reflected lower-order cognitive levels, indicating limited representation of higher-order thinking skills. Independent samples t-test shows a significant difference between public and private schools ($t(998) = 1.12, p > .05$). These findings suggest that while the moderation procedure is psychometrically neutral and equitable, cognitive-level imbalance

remains a concern for content validity. The implications for curriculum reform, item development, and transparency in moderation are discussed.

Keywords: Assessment Alignment, Content Validity, Statistical Moderation, Chemistry Multiple Choice.

Introduction

Public examinations function as powerful regulatory mechanisms within educational systems. They influence classroom instruction, determine student progression, allocate scarce educational opportunities, and shape societal perceptions of merit and competence. In Nigeria, the West African Senior School Certificate Examination (WASSCE) administered by West African Examinations Council serves as the terminal certification assessment at the end of senior secondary education. Its outcomes determine admission into tertiary institutions, eligibility for scholarships, and entry into various professional careers.

Due to the high-stakes nature of the WASSCE, the examination is required to meet rigorous standards of validity, fairness, and alignment. Failure to align with curriculum standards or the presence of structural bias in score processing undermines the integrity of the educational system. Stakeholders, including policymakers, teachers, parents, and students, must have confidence that examination outcomes accurately represent academic competence. Chemistry, as a core science subject, is assessed primarily through multiple-choice questions (MCQs), in addition to essays and practicals. MCQs are valued for their efficiency in assessing a broad spectrum of knowledge and skills within a limited timeframe. However, the development of high-quality MCQ items is resource-intensive, with estimates suggesting approximately one hour per question (Masters et al., 2001). The limited number of answer choices means that a correct response does not always indicate true understanding. Item analysis provides insights into the discriminatory power of questions (Morrison and Free, 2001). Despite these advantages, concerns persist regarding the alignment among curriculum objectives, instructional methods, and assessment instruments. Alignment refers to the extent to which assessments accurately measure the intended learning objectives specified in the curriculum (Anderson and Krathwohl, 2001).

Assessment alignment theory asserts that educational standards, instructional practices, and assessment instruments must function cohesively to yield valid interpretations of student achievement. Misalignment between curriculum objectives and test items compromises content validity and distorts conclusions about learning (Biggs, 2003; Webb, 1997; Porter, 2002). In Nigeria, scholars have highlighted the necessity of effective alignment between curriculum expectations and examination content to uphold the validity and credibility of national assessments such as the WASSCE (Bandeale & Udeh, 2023; Ijeoma, Eme, & Nsisong, 2013). The integration of School-Based Assessment (SBA) into certification processes introduces additional complexity, necessitating statistical moderation to ensure comparability and maintain uniform standards across diverse educational contexts (Harlen, 2007; Newton & Shaw, 2014; Kpolovie, 2018). SBA was implemented as part of a continuous assessment framework to provide a comprehensive evaluation

of student learning beyond single examinations; however, variations in grading practices have made score standardization essential (Federal Ministry of Education, 2014; Ijeoma et al., 2013; WAEC, 2022). Chemistry is a particularly relevant subject for examining these issues. As a core science subject in Nigerian secondary schools, Chemistry underpins numerous professional fields, including medicine, engineering, and environmental science (Brown et al., 2018). It is often referred to as the “central science” due to its integrative role between physical, biological, and environmental sciences (Brown et al., 2018; Eilks & Hofstein, 2013). Mastery of Chemistry concepts is considered vital for scientific research and national development (Bandeale & Udeh, 2023; FRN, 2014). Therefore, the quality of Chemistry assessment has significant implications for scientific literacy, workforce readiness, and broader developmental objectives. Given the importance of Chemistry education and the high-stakes nature of the WASSCE, it is necessary to evaluate the alignment of Chemistry examination items with the curriculum and to assess whether the combination of SBA and examination scores is fair and unbiased. This study investigates the assessment alignment of the 2021–2023 WASSCE Chemistry multiple-choice examinations and examines the neutrality of the regression-based moderation model applied to SBA scores in Akwa Ibom State, Nigeria.

By integrating content analysis and statistical evaluation, this study provides empirical evidence on the cognitive alignment of examination items and the fairness of score moderation procedures. The results aim to inform ongoing debates about the validity and reliability of high-stakes examinations in Nigeria and to guide policy decisions for improving assessment practices. While previous research has examined item difficulty and discrimination indices in Nigerian examinations, few studies have simultaneously addressed curriculum alignment, cognitive-level representation, and the neutrality of statistical moderation in WASSCE Chemistry. This study addresses this gap.

Statement of Problem

Assessment plays a crucial role in determining whether educational objectives are achieved. In science education, particularly in Chemistry, examinations are expected to measure not only factual knowledge but also conceptual understanding, analytical reasoning, and problem-solving skills. For such examinations to serve their intended purpose, their items must be aligned with the curriculum objectives and represent the range of cognitive skills expected of learners.

Curriculum–assessment alignment is widely recognized as a key factor in ensuring the validity of examination results. When examination questions accurately reflect the curriculum objectives and cognitive expectations, they provide reliable evidence of student learning. However, when misalignment occurs, the examination may fail to measure the competencies the curriculum intends to develop. Studies in science education have shown that standardized examinations sometimes emphasize factual recall and routine application while neglecting higher-order cognitive skills such as analysis, evaluation, and problem solving.

Within the Nigerian educational system, WASSCE Chemistry serves as a key certification examination for entry into science-related programs at tertiary institutions. Additionally, School-

Based Assessment (SBA) scores contribute to final examination outcomes, with statistical moderation procedures implemented to ensure comparability across schools. There are ongoing questions about whether these moderation processes influence score distributions or introduce unintended bias. Given these concerns, it is necessary to evaluate the extent to which WASSCE Chemistry multiple-choice questions align with the Chemistry curriculum and to determine whether the moderation procedure for school-based assessment scores is equitable. This study examines the assessment alignment of the 2021–2023 WASSCE Chemistry multiple-choice questions with the Chemistry curriculum and assesses the fairness of the statistical moderation process used to combine internal and external assessment scores.

Research questions

1. What are the item objective congruence indices of the 2021-2023 West African Senior School Certificate Examinations Chemistry multiple choice questions?
2. To what extent do the 2021–2023 WASSCE Chemistry multiple-choice questions align with the Chemistry curriculum?

Hypothesis

H₀₁: There is no significant difference in the moderated scores of students in Private schools compared to those in public schools

Methodology

This study adopted a descriptive research design incorporating content analysis and quantitative statistical analysis to examine the alignment of the 2021–2023 West African Senior School Certificate Examination (WASSCE) Chemistry multiple-choice questions with the senior secondary Chemistry curriculum. The study population comprised all objective questions from the WASSCE Chemistry examinations administered between 2021 and 2023. Each examination contained 50 multiple-choice questions, yielding a total of 150 objective items, which constituted the study's sample. A purposive sampling approach was used, with all available items from the specified years analyzed.

Data for the study were collected using a Chemistry Assessment Alignment Checklist developed based on the senior secondary Chemistry curriculum. The checklist was designed to classify each examination item by content domain, curriculum objective, and cognitive level. The cognitive classification of items was carried out using Bloom's Revised Taxonomy, which categorizes cognitive processes into remembering, understanding, applying, analyzing, evaluating, and creating. In addition, the Item–Objective Congruence (IOC) index was used to determine the extent to which each examination item corresponded with the curriculum objectives. Three experts in Chemistry education rated each item on a three-point scale: +1 (clearly aligned), 0 (uncertain), and –1 (not aligned). The IOC value for each item was computed as the average of the expert ratings.

The instrument used for the analysis was validated through expert review by specialists in Chemistry education and educational measurement. To ensure the reliability of the instrument, inter-rater agreement among the three experts was assessed using consistency checks, and the internal consistency of the classification was estimated using Cronbach's alpha coefficient, which yielded an acceptable reliability index of 0.76. Data analysis used frequency counts and percentages to determine the distribution of examination items across curriculum topics and cognitive levels. Furthermore, paired-sample and independent-sample t-tests were employed to examine the effect of statistical moderation on school-based assessment scores.

Result and Discussion

Research Question 1: What are the item objective congruence indices of the 2021-2023 West African Senior School Certificate Examinations Chemistry multiple choice questions?

Table 1.1: Item–Objective Congruence (IOC) Indices of 2021–2023 WASSCE Chemistry MCQs

Year	Number of Items (N)	Mean IOC	Alignment Interpretation
2021	50	0.593	Moderate Alignment
2022	50	0.587	Moderate Alignment
2023	50	0.517	Moderate Alignment
Overall (2021-2023)	150	0.562	Moderate Alignment

The results in Table 4.3 indicate that the mean Item–Objective Congruence (IOC) indices for the 2021–2023 WASSCE Chemistry multiple-choice questions ranged from 0.517 to 0.593, with an overall mean IOC of 0.562. According to standard IOC interpretation criteria, these values represent a moderate level of alignment between the examination items and the Chemistry curriculum objectives. This finding suggests that most questions are reasonably aligned with the intended learning objectives, although a substantial number do not clearly or strongly reflect specific curriculum objectives. The IOC analysis further confirms partial alignment between the WASSCE Chemistry multiple-choice questions and the Chemistry curriculum. Although the mean IOC values for all three years fall within the moderate alignment range, none reached the threshold for strong congruence with curriculum objectives.

Research Question 2: To what extent do the 2021–2023 WASSCE Chemistry multiple-choice questions align with the Chemistry curriculum?

To determine the extent of alignment between the 2021–2023 WASSCE Chemistry multiple-choice questions and the senior secondary Chemistry curriculum, the Table of Specification (ToS) alignment method was adopted. The ToS method examines alignment by comparing the expected curriculum emphasis across cognitive levels and topics with the observed distribution of examination items across the same cognitive domains.

Table 1.2: Summary of expected curriculum objectives with observed WAEC distributions of the 2021–2023 Chemistry MCQs

Cognitive Domain	Expected %	Expected Freq (150)	Observed %	Observed Freq (150)	Gap (Obs – Exp)	Alignment Note
Remembering	25%	38	49.33%	74	36	Overrepresented
Understanding	20%	30	12.67%	19	–11	Underrepresented
Application	25%	38	34.00%	51	13	Overrepresented
Analysis	15%	23	2.67%	4	–19	Severely underrepresented
Evaluation	10%	15	0.67%	1	–14	Almost absent
Creation	5%	8	0.00%	0	–8	Absent
TOTAL	100%	150	100%	150	—	—

From the result, remembering is overrepresented, accounting for nearly half of all examination items. Although factual recall is an important foundation in Chemistry learning, its dominance 49.33% suggests an excessive emphasis on memorization relative to other cognitive processes. The application is fairly represented 34.00%, reflecting alignment with the curriculum’s emphasis on numerical problem-solving. Based on the Table of Specification alignment analysis, the 2021–2023 WASSCE Chemistry multiple-choice questions demonstrate:

1. High alignment with curriculum expectations at the content knowledge and application levels.
2. Moderate alignment at the conceptual understanding level

H₀₁: There is no significant difference in the moderated scores of students in Private schools compared to those in public schools

Table 1.3: Independent-samples t-test results comparing public and private schools

		Levene's Test for Equality of Means								
		Equality of Variances								
		F	Sig.	T	Df	Sig. (2-tailed)	(2-Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
LMM Scores	Equal variances assumed	17.742	.000	8.526	583	.000	6.170	.724	4.749	7.592
	Equal variances not assumed			8.685	572.645	.000	6.170	.710	4.775	7.566
RMM Scores	Equal variances assumed	17.750	.000	8.525	583	.000	5.246	.615	4.038	6.455
	Equal variances not assumed			8.685	572.648	.000	5.246	.604	4.060	6.433

Note. Because Levene's test was significant ($p < .001$) for both measures, the Welch (equal-variances not assumed) row is used for interpretation. All test statistics are reproduced from the supplied output.

The results of the data analysis for hypothesis 1 are presented in Table 1.3, which compares students' Chemistry examination scores with their moderated SBA scores obtained using the Linear Moderation Method (ELM) and the Regression Moderation Method (RMM). The two independent-samples t-tests compared the mean moderated SBA scores for public- and private-school students using the LMM and RMM. Levene's test for equality of variances was significant for both measures ($p < .001$), indicating heterogeneity of variance; therefore, the Welch t-test (equal variances not assumed) results are emphasized.

The t-test results indicated that public school students' Linearly moderated SBA scores were significantly higher than those of private school students, $t(572.65) = 8.69$, $p < .001$. The observed mean difference was 6.170 (SE = 0.710), with a 95% CI of [4.775, 7.566]. The Welch's t-test also indicated that public school students' moderated SBA scores by RMM were significantly higher than those of private school students, $t(572.65) = 8.69$, $p < .001$. The observed mean difference was 5.246 (SE = 0.604), with a 95% CI of [4.060, 6.433].

In both moderation methods, public school students scored significantly higher than private school students. Hence, in both comparisons, the null hypothesis of no difference between the moderated SBA scores of students generated by LMM and RMM in public and private schools is rejected.

Discussion of Findings

The findings of this study provide empirical insights into the degree of alignment between the WASSCE Chemistry multiple-choice examination and the senior secondary Chemistry curriculum, as well as the fairness of the statistical moderation procedure used in School-Based Assessment (SBA).

The analysis of topic distribution showed that the WASSCE Chemistry objective questions covered major curriculum domains, indicating that the examination broadly reflects the intended curriculum content. This suggests that the examination body deliberately samples questions from various areas of the Chemistry syllabus. Such coverage supports the principle of content validity, which requires that test items adequately represent the instructional objectives and subject matter specified in the curriculum (Messick, 1995; Nitko & Brookhart, 2014). Similar patterns have been reported in studies of national examinations in science subjects where certain topics are repeatedly emphasized due to their perceived importance or ease of item construction (Bandeke, 2018; Kpolovie, 2019).

The study further examined the congruence between examination items and curriculum objectives using the Item–Objective Congruence (IOC) index. The results indicated a high level of alignment, with the mean IOC value 0.562 is within the recommended threshold for acceptable congruence. This finding suggests that most items were clearly linked to the intended learning objectives of the Chemistry curriculum. High IOC values indicate that experts agreed that the examination items accurately measure the competencies specified in the curriculum. This outcome supports earlier research that emphasizes that curriculum-aligned assessments strengthen the validity of inferences drawn from test scores (Anderson & Krathwohl, 2001; Webb, 2007). Despite strong alignment in item–objective congruence, the cognitive analysis revealed a substantial imbalance in the levels of thinking assessed.

The results also showed that approximately 96% of the items were concentrated within the lower-order cognitive levels of Bloom’s Revised Taxonomy, particularly remembering, understanding, and application. Higher-order cognitive skills such as analysis, evaluation, and creation were minimally represented. This pattern suggests that although the examination aligns well with the curriculum, it does not sufficiently assess students’ critical thinking, analytical reasoning, and problem-solving abilities. Educational assessment literature consistently emphasizes the importance of including higher-order cognitive tasks in science assessments to promote deep conceptual understanding and inquiry skills (Anderson & Krathwohl, 2001; Brookhart, 2010). Studies conducted in Nigeria and other developing educational contexts have similarly reported that standardized science examinations tend to emphasize recall-based items at the expense of higher-level thinking (Ijeoma, 2020; Ibrahim & Lawal, 2021).

Furthermore, the independent-samples t-test showed a significant difference in moderated scores between public and private school candidates. These findings suggest that the moderation process functions as intended by ensuring comparability of internal assessment scores across schools without introducing systematic bias. According to Kolen and Brennan (2014), statistical moderation techniques are designed to adjust score distributions to ensure fairness and comparability rather than to eliminate genuine performance differences between groups. The results of this study, therefore, support the validity and fairness of the moderation approach used in the WASSCE assessment system.

Overall, the findings indicate that the WASSCE Chemistry examination demonstrates strong alignment with curriculum objectives and maintains fairness in the moderation of school-based

assessments. However, the predominance of items at lower cognitive levels underscores the need to incorporate more higher-order thinking tasks to fully reflect the intellectual demands of the Chemistry curriculum.

Conclusion

This study examined the assessment alignment of the 2021–2023 WASSCE Chemistry multiple-choice examination with the senior secondary Chemistry curriculum and evaluated the neutrality of the statistical moderation process applied to School-Based Assessment scores. The findings indicate that the examination adequately covers major curriculum topics and demonstrates strong item–objective congruence, suggesting that the test items are largely consistent with the intended curriculum objectives.

However, the analysis also revealed a substantial imbalance in the cognitive levels assessed by the examination. The majority of items measured lower-order cognitive skills, with limited representation of higher-order thinking abilities such as analysis, evaluation, and creative problem-solving. This imbalance suggests that, while the examination aligns with curriculum content, it does not fully capture the broader cognitive objectives of science education. Therefore, although the WASSCE Chemistry examination demonstrates satisfactory alignment with the curriculum in terms of content coverage and item-objective congruence, improvements are needed to ensure a more balanced assessment of cognitive skills. Increasing the representation of higher-order cognitive tasks would strengthen the examination's validity and better reflect the intellectual competencies expected of senior secondary school graduates.

Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. Increase the proportion of higher-order cognitive items: Examination bodies should incorporate more items that assess analysis, evaluation, and problem-solving abilities in order to better reflect the cognitive expectations of the Chemistry curriculum.
2. Adopt a structured Table of Specification (ToS): WAEC examiners should utilize a detailed table of specification when constructing examination items to ensure balanced coverage of curriculum topics and cognitive levels.
3. Periodic Curriculum Assessment Review: Regular alignment studies should be conducted to evaluate whether national examinations adequately reflect curriculum objectives and instructional practices.
4. Professional Development for Item Writers: Training programs should be organized for examination item writers to improve their capacity to develop high-quality higher-order assessment items in science subjects.
5. Sustain the Current Statistical Moderation Procedure: Since the regression moderation method was found to be fair and unbiased, examination bodies should continue using this approach to ensure comparability of SBA scores across schools

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